

Problem 2: Bag-of-Words and Document Distances

Problem 2: Bag-of-Words and Document Distances (65 points)

Dataset: NYT art vs music corpus

Task 1 — Data check (4 pts)

```
Resolved DATA_DIR: ~/Desktop/data
DATA_DIR exists: True
Count of XML files in art/: 57
Count of XML files in music/: 45
First 5 doc IDs in art/: ['0068412', '0142012', '0256288',
'0332083', '0365256']
First 5 doc IDs in music/: ['0023931', '0075170', '0170797',
'0178569', '0299849']
```

Task 2 — Tokenization preview (6 pts)

```
Number of docs in art_tokens: 57
Number of docs in music_tokens: 45

First 20 tokens of art doc 1595645:
['red', 'grooms', 'by', 'arthur', 'c', 'danto', 'timothy', 'hyman',
'and',
'marco', 'livingstone', 'illustrated', 'pp', 'rizzoli', 'red',
'grooms',
'so', 'perfectly', 'captures', 'new']
```

Preprocessing. `strip_text` lowercases the text and removes apostrophes, maps all digit runs to #, strips remaining punctuation to spaces, and splits on whitespace — producing a clean list of lowercase word/# tokens with no punctuation.

Task 3 — Targeted lookups (15 pts)

For art document **1595645**:

- (a) 37th token (index 36): experiencing
- (b) Count of 'the' via loop/sum: 103
 Count of 'the' via Counter: 103

Both methods agree (103), as expected — same tally, two implementations.

Task 4 — Build BoW and variants (15 pts)

X_{bow} shape (docs x terms): (102, 4432)

Label counts:

```
art      57
music    45
```

Sample of X_{raw} (first 5 docs, first 6 terms):

	lead	artists	thinking	about	au	in
0068412	2	17	2	10	2	54
0142012	1	6	0	0	0	17
0256288	1	0	0	0	0	33
0332083	1	1	0	2	0	30
0365256	1	0	0	0	0	20

102 documents (57 art + 45 music), 4,432 terms after dropping singleton words.

(a) Normalization. Row-normalizing (L1 or L2) removes the effect of document length, so distances reflect relative word usage rather than raw article size.

(b) IDF weighting. IDF down-weights terms common across most documents (e.g., “the,” “said”) and up-weights rare, topic-specific terms (e.g., “gallery,” “orchestra”), since $\text{idf}_j = \log(n/df_j)$. This shifts distance computation toward informative vocabulary.

Task 5 — Distance matrices and analysis (25 pts)

measure	avg_within	avg_between	gap (between – within)	gap as % of avg_within
raw	75.981	76.112	0.132	0.17%
norm_sum	0.099	0.103	0.004	3.81%
norm_l2	0.731	0.755	0.024	3.34%
raw_idf	89.799	88.375	–1.424	–1.59%
norm_sum_idf	0.129	0.133	0.003	2.56%
norm_l2_idf	1.373	1.383	0.010	0.74%

Interpretation. All representations show a positive within/between gap except raw_idf, where it flips negative. This is because unnormalized counts are dominated by document length (41–2,615 tokens, std $\approx$ 463), so same-topic articles of very different lengths can still look far apart; IDF only amplifies this length effect rather than correcting it.

Once length is controlled by normalization, separation improves: `norm_sum` gives the largest relative gap (3.8%), followed by `norm_l2` (3.3%). Adding IDF on top of normalization doesn't help further — likely because art and music articles from the NYT culture desk share a lot of “art-world” vocabulary, and IDF's aggressive down-weighting of corpus-wide common terms ends up suppressing some of that still-useful shared vocabulary along with true noise words.

Overall: length normalization matters far more than IDF weighting for this corpus. `norm_sum` (L1-normalized raw counts) gives the best art-vs-music separation among the six representations.